

## MOSS VALE, Australia

WMO: 947460

Lat: 34.5253S

Lon: 150.4217E

Elev: 2228

StdP: 13.55

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 29.3       | 31.9 | 22.3                          | 18.2 | 52.5 | 25.1 | 20.8 | 48.0 | 33.8                  | 46.5 | 31.5 | 46.1 | 2.3                   | 250  | 0.613 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 19.5                   | 90.1            | 63.9 | 85.4 | 63.2 | 81.4 | 62.6 | 69.1                | 78.7 | 67.5 | 76.8 | 66.1 | 74.9 | 14.9                 | 290  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.6                            | 106.5 | 71.8 | 64.9 | 100.2 | 70.2 | 63.5 | 95.1 | 68.8 | 34.6          | 78.9 | 33.3 | 77.1 | 32.1 | 74.9 | 75.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 30.9          | 26.7 | 23.3 | DB                         | 25.0 | 98.1               | 1.4 | 3.7                                                | 24.1 | 100.7      | 23.3 | 102.9      | 22.5 | 105.0      | 21.5 | 107.8 | (4) |
| (5)               |      |      | WB                         | 24.2 | 72.4               | 1.4 | 2.3                                                | 23.2 | 74.1       | 22.4 | 75.4       | 21.6 | 76.7       | 20.6 | 78.4  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 55.8   | 67.4  | 65.5  | 62.1  | 56.5  | 50.0  | 46.0  | 44.7  | 46.3  | 51.2  | 55.9  | 60.3  | 63.9  | (6)  |  |
|                                                                        | DBStd       | 9.54   | 6.64  | 6.02  | 5.51  | 5.31  | 4.46  | 3.94  | 4.07  | 4.56  | 5.63  | 6.42  | 6.84  | 6.63  | (7)  |  |
|                                                                        | HDD50       | 568    | 0     | 0     | 0     | 6     | 56    | 131   | 171   | 136   | 49    | 16    | 3     | 0     | (8)  |  |
|                                                                        | HDD65       | 3719   | 52    | 61    | 124   | 261   | 465   | 571   | 628   | 581   | 414   | 289   | 172   | 101   | (9)  |  |
|                                                                        | CDD50       | 2670   | 538   | 433   | 375   | 200   | 56    | 11    | 8     | 20    | 86    | 200   | 313   | 430   | (10) |  |
|                                                                        | CDD65       | 344    | 125   | 74    | 34    | 5     | 0     | 0     | 0     | 0     | 1     | 7     | 32    | 66    | (11) |  |
|                                                                        | CDH74       | 3237   | 1187  | 540   | 210   | 48    | 0     | 0     | 0     | 0     | 14    | 129   | 408   | 701   | (12) |  |
|                                                                        | CDH80       | 1162   | 514   | 203   | 38    | 7     | 0     | 0     | 0     | 0     | 2     | 19    | 138   | 241   | (13) |  |
| (14) Wind                                                              | WSAvg       | 10.9   | 11.2  | 10.7  | 10.1  | 8.9   | 10.0  | 10.5  | 11.5  | 11.7  | 12.1  | 11.6  | 11.4  | 11.4  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 36.5   | 3.3   | 3.6   | 3.7   | 3.3   | 2.9   | 3.1   | 1.9   | 2.6   | 2.2   | 3.0   | 3.6   | 2.8   | (15) |  |
|                                                                        | PrecMax     | 59.7   | 8.7   | 12.2  | 18.3  | 17.0  | 9.1   | 13.5  | 6.1   | 13.8  | 8.6   | 8.9   | 12.8  | 7.7   | (16) |  |
|                                                                        | PrecMin     | 19.4   | 0.5   | 0.1   | 0.2   | 0.1   | 0.1   | 0.0   | 0.1   | 0.1   | 0.2   | 0.2   | 0.1   | 0.0   | (17) |  |
|                                                                        | PrecStd     | 10.6   | 2.1   | 2.8   | 3.1   | 3.4   | 2.3   | 3.1   | 1.5   | 3.2   | 1.7   | 2.2   | 2.7   | 1.9   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 96.9  | 93.5  | 84.2  | 80.2  | 69.1  | 61.9  | 62.1  | 66.2  | 75.4  | 82.6  | 90.8  | 92.8  | (19) |  |
|                                                                        |             | MCWB   | 65.7  | 66.3  | 63.5  | 59.9  | 54.7  | 51.3  | 50.9  | 51.0  | 54.8  | 57.7  | 61.8  | 63.7  | (20) |  |
|                                                                        | 2%          | DB     | 91.1  | 85.9  | 79.6  | 73.7  | 65.1  | 58.1  | 57.8  | 61.7  | 71.1  | 77.5  | 83.7  | 86.5  | (21) |  |
|                                                                        |             | MCWB   | 65.1  | 63.8  | 62.6  | 58.0  | 53.4  | 50.3  | 48.2  | 49.6  | 53.7  | 57.1  | 60.3  | 62.7  | (22) |  |
|                                                                        | 5%          | DB     | 86.1  | 80.4  | 75.5  | 69.3  | 62.3  | 55.5  | 54.9  | 58.8  | 66.4  | 73.5  | 78.2  | 81.7  | (23) |  |
|                                                                        |             | MCWB   | 64.6  | 64.5  | 63.0  | 57.7  | 52.7  | 49.2  | 47.0  | 48.3  | 52.4  | 55.5  | 60.0  | 61.9  | (24) |  |
|                                                                        | 10%         | DB     | 80.5  | 75.5  | 71.5  | 66.0  | 59.3  | 53.2  | 52.6  | 55.8  | 62.5  | 69.2  | 73.5  | 77.2  | (25) |  |
|                                                                        |             | MCWB   | 64.3  | 63.9  | 62.0  | 56.9  | 51.6  | 48.2  | 46.0  | 46.8  | 50.6  | 55.0  | 60.1  | 61.0  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 71.8  | 71.7  | 68.4  | 63.6  | 59.1  | 54.9  | 55.0  | 54.8  | 58.5  | 62.9  | 67.6  | 68.4  | (27) |  |
|                                                                        |             | MCDB   | 83.2  | 80.1  | 75.9  | 70.9  | 63.0  | 58.0  | 57.3  | 61.9  | 68.7  | 71.7  | 75.2  | 78.6  | (28) |  |
|                                                                        | 2%          | WB     | 69.4  | 69.0  | 66.5  | 61.2  | 56.2  | 52.2  | 51.1  | 51.7  | 56.1  | 60.3  | 64.8  | 66.3  | (29) |  |
|                                                                        |             | MCDB   | 80.4  | 77.5  | 73.5  | 67.7  | 61.2  | 55.6  | 54.8  | 58.4  | 65.2  | 69.8  | 75.2  | 76.9  | (30) |  |
|                                                                        | 5%          | WB     | 67.9  | 67.1  | 64.7  | 59.8  | 54.2  | 50.9  | 48.8  | 50.0  | 54.5  | 58.6  | 62.9  | 64.6  | (31) |  |
|                                                                        |             | MCDB   | 77.8  | 75.6  | 71.0  | 65.8  | 59.8  | 54.2  | 52.4  | 56.1  | 62.4  | 68.6  | 73.3  | 75.1  | (32) |  |
|                                                                        | 10%         | WB     | 66.4  | 65.5  | 63.3  | 58.5  | 52.8  | 49.4  | 47.0  | 48.1  | 52.7  | 56.8  | 61.2  | 63.2  | (33) |  |
|                                                                        |             | MCDB   | 75.9  | 72.7  | 69.2  | 64.0  | 58.1  | 52.7  | 51.1  | 54.2  | 60.3  | 66.4  | 70.5  | 73.0  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 19.5  | 16.4  | 15.8  | 16.7  | 17.2  | 14.0  | 15.0  | 16.8  | 18.4  | 19.2  | 18.9  | 19.6  | (35) |  |
|                                                                        |             | MCDBR  | 30.8  | 25.7  | 22.5  | 22.0  | 22.4  | 18.5  | 18.6  | 21.8  | 25.5  | 27.0  | 28.7  | 28.5  | (36) |  |
|                                                                        |             | MCWBR  | 9.5   | 8.6   | 8.8   | 9.6   | 12.4  | 12.4  | 11.1  | 11.7  | 11.8  | 11.0  | 10.0  | 9.3   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 23.3  | 20.3  | 18.7  | 16.7  | 18.6  | 14.3  | 14.2  | 18.0  | 20.7  | 22.6  | 23.1  | 22.3  | (38) |  |
|                                                                        |             | MCWBR  | 9.7   | 8.8   | 8.5   | 8.4   | 11.5  | 10.5  | 10.1  | 10.9  | 11.2  | 11.0  | 10.2  | 9.8   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.357 | 0.349 | 0.326 | 0.302 | 0.288 | 0.278 | 0.271 | 0.279 | 0.305 | 0.321 | 0.345 | 0.351 | (40) |  |
|                                                                        | taud        |        | 2.492 | 2.543 | 2.583 | 2.630 | 2.622 | 2.656 | 2.670 | 2.634 | 2.551 | 2.529 | 2.492 | 2.491 | (41) |  |
|                                                                        | Ebn at Noon |        | 311   | 307   | 302   | 293   | 281   | 276   | 285   | 298   | 305   | 312   | 312   | 314   | (42) |  |
|                                                                        | Edh at Noon |        | 36    | 33    | 30    | 25    | 22    | 20    | 21    | 24    | 30    | 33    | 36    | 37    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2040  | 1720  | 1478  | 1203  | 955   | 745   | 869   | 1151  | 1546  | 1810  | 1907  | 2093  | (44) |  |
|                                                                        | RadStd      |        | 163   | 162   | 131   | 107   | 58    | 64    | 67    | 75    | 100   | 164   | 175   | 188   | (45) |  |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |      |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |      |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m)  |      |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | N/A  | (46) |
| (47) | Regional (36 neighbors) | +0.77   | N/A    | N/A     | +1.37 | N/A   | N/A         | -2    | -153  | +328  | +167 | (47) |

Nomenclature: See separate page